ITA03-MP Assignment 1

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MOBILE COMPUTING

ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Nature of Assignment: Individual
Duration: 1 Week
Total Marks: 50
Submission Date: _____

Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township
(Total Marks – 50)

Duration:1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon).
- Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.

- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km² stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks

Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

Evidence Expected

Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

TASK 2 – Identify the Governing Design Parameter **2 Marks**

Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km².

- Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour **4 Marks**

Guiding Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each blocking behaviour.
- analyse how shadowing and a changing path-loss exponent shift each case.
- Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

TASK 5 – Scaling and Sensitivity Analysis **3 Marks**

Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical?

Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto$ 1/N, SIR $\propto$ N^(n/2), cells $\propto$ $1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting **3 Marks**

Guiding Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round k − 1?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius R → R/2).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure.
- Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

TASK 7 – Solve the Recurrence and Interpret the Result **4 Marks**

Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4(D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

TASK 8 – Implement and Test the Planning Strategies **4 Marks**

Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator.
- Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

TASK 9 – Record and Present Experimental Performance **3 Marks**

Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

Evidence Expected

Systematically collected data presented in comparable tables and graphs.

TASK 10 – Compare Theory with Simulation **4 Marks**

Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy.
- Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

Evidence Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

TASK 11 – Recommend and Defend the Final Design **4 Marks**

Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.
- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design **8 Marks**

Changed Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions.

You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis – 2 Marks

Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density.
- Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

B. Evaluation of the Original Design – 2 Marks

Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.
- Assess whether the design can be reused unchanged and what fails first.
- Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge.

C. Adaptation or Redesign – 2 Marks

Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing).
- Describe the revised strategy and what changes in reuse pattern, power and handoffs.

- Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect.

D. Evidence-Based Defence – 2 Marks

Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

MARK DISTRIBUTION

Task	**Marks**
Task 1 – Design Alternative Frequency-Planning Strategies	5
Task 2 – Identify the Governing Design Parameter	2
Task 3 – Derive the Capacity and Interference Functions Mathematically	6
Task 4 – Analyse Best, Average and Worst-Case Behaviour	4
Task 5 – Scaling and Sensitivity Analysis	3
Task 6 – Formulate the Recurrence for Repeated Cell Splitting	3
Task 7 – Solve the Recurrence and Interpret the Result	4

Task 8 – Implement and Test the Planning Strategies	4
Task 9 – Record and Present Experimental Performance	3
Task 10 – Compare Theory with Simulation	4
Task 11 – Recommend and Defend the Final Design	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	8
TOTAL	**50**

ANSWERS – MOBILE COMPUTING ASSIGNMENT 1 (CO1)

Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township

GIVEN DATA

Parameter	Value
Township area	12 km²
Available duplex traffic channels	336
Subscriber forecast	1,000; 10,000; 20,000; 100,000+
Busy-hour traffic per subscriber	0.025 Erlang
Maximum blocking probability	2%
Acceptable SIR range	12–18 dB
Path-loss exponent	2.5–4
Stadium area	1 km²
Event attendees	25,000
Event traffic per attendee	0.05 Erlang
Event duration and frequency	4 hours; approximately twice monthly

Total offered traffic is A_total = U × 0.025. Therefore, 1,000 subscribers generate 25 E, 10,000 generate 250 E, 20,000 generate 500 E, and 100,000 generate 2,500 E.

TASK 1 – DESIGN ALTERNATIVE FREQUENCY-PLANNING STRATEGIES (5 MARKS)

Frequency planning shares the fixed allocation of 336 duplex traffic channels among cells while controlling co-channel interference. Three distinct strategies are considered.

Strategy 1: Fixed Frequency Reuse

Principle: spatial reuse. The 336 channels are divided into N groups, with each group assigned to a different cell in a cluster. The same group is reused in another cluster after sufficient geographical separation.

For N = 7, channels per cell = 336/7 = 48.

- Advantages: simple allocation, predictable interference, low implementation complexity and suitability for relatively uniform traffic.
- Disadvantages: channels may be unused in low-traffic cells, overloaded cells cannot easily borrow capacity, and small N may increase interference.

- Access: in an FDMA/TDMA system, channels or time slots are divided among users. Control channels carry signalling, registration and call setup; traffic channels carry voice/data.

Iterative procedure: generate candidate cluster sizes $N = i^2 + ij + j^2$; calculate channels per cell, reuse distance and SIR; reject candidates below the SIR threshold; select a feasible N; then assign channel groups across the hexagonal grid.

Pseudocode:

```
channels = 336
for i in valid_integer_range:
    for j in valid_integer_range:
        N = i*i + i*j + j*j
        if N is valid:
            channels_per_cell = channels // N
            D = R * sqrt(3*N)
            calculate SIR
            if SIR >= 12 dB:
                accept N and assign frequency groups
```

Strategy 2: Sectorised Frequency Reuse

Principle: spatial filtering. Each base-station cell is divided into directional sectors, commonly three 120-degree sectors. Directional antennas serve users in a restricted angular region and reduce the number of dominant co-channel interferers.

- Advantages: improves SIR, supports higher traffic density and can increase reuse efficiency.
- Disadvantages: requires additional antennas/radio equipment, careful antenna alignment and more maintenance.
- Access: frequency/time resources are assigned to sectors; control and traffic channels are managed within each sector.

Procedure: divide the cell into sectors; install directional antennas; assign sector channel groups; estimate sector-specific interference; if SIR is inadequate, adjust antenna direction, power or reuse planning; otherwise accept the plan.

Strategy 3: Dynamic Channel Assignment and Cell Splitting

Principle: trunking efficiency and hierarchical coverage. Dynamic channel assignment allocates channels according to current demand. A congested cell may borrow an available channel from a neighbour if interference constraints are satisfied. If congestion persists, the cell can be split into smaller cells.

- Advantages: efficient use of channels, adaptation to uneven traffic, and suitability for crowded locations and special events.
- Disadvantages: more complex control, additional handoffs and possible infrastructure/backhaul costs.
- Access: channels are selected dynamically; control signalling coordinates borrowing, allocation and handoff.

Procedure: measure cell traffic; if the cell is not overloaded, continue normal operation; otherwise check neighbour channels; borrow a safe channel if available; if not, split the overloaded cell; then monitor traffic, SIR and handoffs.

Preprocessing and cost

- Prepare a traffic-demand map for academic blocks, hostels, residential areas, roads and stadium.
- Conduct propagation surveys to estimate path loss, shadowing and interference.
- Generate a hexagonal grid and identify candidate base-station sites.
- Estimate site acquisition, survey, equipment, backhaul, power and maintenance costs.

The operator bears the planning, survey, installation and maintenance costs. The university may assist with land access and site information.

Conclusion: fixed reuse provides a simple baseline, sectorisation improves interference performance, and dynamic assignment with selective splitting provides flexibility as demand grows.

TASK 2 – IDENTIFY THE GOVERNING DESIGN PARAMETER (2 MARKS)

For fixed reuse, the dominant parameter is cluster size N. It controls the channels per cell, $C = 336/N$, and the reuse distance $D = R\sqrt{(3N)}$.

For sectorised reuse, the dominant interference parameter is the number of dominant co-channel interferers I. Directional antennas reduce the interfering power received in a sector.

For dynamic assignment and cell splitting, the dominant geometric parameter is cell radius R, together with the availability of channels in neighbouring cells. The number of cells is proportional to $1/R^2$.

Subscribers	Traffic	Planning response
1,000	25 E	Basic fixed reuse may be sufficient under uniform-load assumptions.
10,000	250 E	Monitor busy cells and introduce sectorisation where needed.
20,000	500 E	Add capacity using smaller cells and dynamic assignment.
100,000+	2,500 E	Use dense small cells, sectorisation and traffic management.

Conclusion: N governs fixed reuse, I governs sectorised interference, and R governs cell splitting. These parameters must be adjusted as subscriber demand increases.

TASK 3 – DERIVE CAPACITY AND INTERFERENCE FUNCTIONS MATHEMATICALLY (6 MARKS)

3.1 Cluster-size relation

For a regular hexagonal cellular grid, valid cluster sizes are $N = i^2 + ij + j^2$, where i and j are non-negative integers describing movement between co-channel cells. Choosing $i = 2$ and $j = 1$ gives $N = 2^2 + (2)(1) + 1^2 = 4 + 2 + 1 = 7$.

3.2 Reuse distance and reuse ratio

The reuse distance is $D = R\sqrt{(3N)}$. For $R = 0.5$ km and $N = 7$: $D = 0.5\sqrt{21} = 2.291$ km. The reuse ratio is $Q = D/R = 2.291/0.5 = 4.583$.

3.3 Hexagonal cell area and number of cells

$A_cell = (3\sqrt{3}/2)R^2$. With $R = 0.5$ km, $A_cell = 2.598 \times 0.25 = 0.6495$ km². The ideal number of cells is $M = 12/0.6495 = 18.47$, rounded upward to 19 cells. Real deployments may need additional sites due to terrain, buildings, coverage overlap and road geometry.

3.4 Channels per cell

Channels per cell = $S/N = 336/7 = 48$ duplex traffic channels per cell.

3.5 Erlang-B traffic capacity

Erlang-B blocking is B(C,A) = (A^C/C!)/Σ[k=0 to C](A^k/k!), where C is the number of channels and A is offered traffic in Erlangs. For C = 48 and target B = 0.02, numerical solution gives approximately 38.39 Erlangs per cell.

For 19 ideal cells, total capacity is approximately 19 × 38.39 = 729.41 Erlangs. Subscriber capacity is 729.41/0.025 ≈ 29,176 subscribers. This assumes uniform independent cell loading and the same channel capacity in each cell.

3.6 Co-channel SIR

Using I dominant first-tier interferers at approximately equal distance, S/I ≈ (D/R)^n/I. Since D/R = √(3N), SIR ≈ (3N)^(n/2)/I. In decibels: SIR_dB = 10 log10[(3N)^(n/2)/I].

For N = 7, n = 4 and I = 6: SIR = 10 log10(21^2/6) = 10 log10(73.5) = 18.66 dB, approximately 18.7 dB.

For n = 2.5: SIR = 10 log10(21^1.25/6) ≈ 8.74 dB, below the 12 dB target.

3.7 Sectorisation effect

In an idealised three-sector example, assume the dominant interferer count falls from six to two. The gain is 10 log10(6/2) = 4.77 dB. Thus the n = 2.5 estimate becomes 8.74 + 4.77 = 13.51 dB. Actual performance depends on antenna beamwidth, side lobes, front-to-back ratio and geometry.

3.8 Cost-capacity relationship

A simple cost model is C_total = C_survey + M·C_site + C_equipment + C_maintenance. Cost per supported Erlang is C_per_Erlang = C_total/A_system. Fixed reuse has lower planning complexity; sectorisation adds antenna cost; dynamic assignment and splitting add management and site costs but provide greater flexibility.

Conclusion: cluster size controls reuse, cell radius controls site count, and path loss/interferer geometry control SIR. Sectorisation is required where the open-campus propagation estimate falls below the target.

TASK 4 – BEST, AVERAGE AND WORST-CASE BEHAVIOUR (4 MARKS)

4.1 Best-case SIR

Best SIR generally occurs close to the serving base station, where desired received power is high and interfering signals are relatively weaker. Illustrative example: desired power = 100 units and interference = 2 units. SIR = 100/2 = 50; SIR_dB = 10 log10(50) = 16.99 dB.

4.2 Average-case SIR

Average SIR generally occurs at a moderate distance from the serving base station. Illustrative example: desired power = 40 units and interference = 4 units. SIR = 40/4 = 10, or 10 dB.

4.3 Worst-case SIR

Worst SIR generally occurs near a cell boundary or corner. The serving base station is farther away and several co-channel cells may be at comparable distances. Illustrative example: desired power = 10 units and interference = 5 units. SIR = 2, or 3.01 dB.

These three received-power examples illustrate the calculation only; they are not measured values from the township.

4.4 Traffic and blocking

Traffic period	Expected load	Blocking behaviour
Off-peak	Low	Lowest blocking
Average period	Moderate	Low to moderate blocking
Busy hour	High	Higher blocking
Stadium event	Very high and concentrated	Potentially highest blocking

At 48 channels, the approximate offered traffic at 2% blocking is 38.39 Erlangs. If offered traffic exceeds this value, blocking rises above the target.

4.5 Shadowing and path loss

Shadowing from buildings and trees can weaken the serving signal, create local dead zones and change SIR. Received power follows $P_r \propto 1/d^n$. An open campus may use n = 2.5, a moderately obstructed area n = 3, and a dense built-up area n = 4. A larger n attenuates distant interferers more quickly, although obstacles may also weaken the desired signal.

Conclusion: best SIR is usually near the serving base station; worst SIR is near cell boundaries. Blocking is lowest off-peak and highest during busy hours or events.

TASK 5 – SCALING AND SENSITIVITY ANALYSIS (3 MARKS)

Channels per cell C = S/N, so C ∝ 1/N. The approximate SIR is proportional to (3N)^(n/2), so SIR ∝ N^(n/2). The hexagonal cell area is proportional to R^2; therefore M ∝ $1/R^2$.

If radius is halved, the ideal number of cells increases by a factor of four. This improves spatial capacity opportunities but increases site and handoff requirements.

Path-loss exponent n	Approximate SIR for N=7, I=6
2.5	8.74 dB
3	12.17 dB
4	18.66 dB

The initial idealised system capacity is about 729.41 Erlangs, equivalent to approximately 29,176 subscribers at 0.025 Erlang each. Above this approximate break-even point, capacity enhancement is required. At 100,000 subscribers, demand is 2,500 Erlangs, so the initial design is insufficient.

Conclusion: capacity decreases with N, SIR increases with N, and site count increases as R decreases. Sectorisation and selective splitting become important as demand grows.

TASK 6 – RECURRENCE FOR REPEATED CELL SPLITTING (3 MARKS)

Let M_0 be the initial number of cells, R_0 the initial radius, C_0 the initial capacity and P_0 the initial transmit power. Each uniform split replaces one cell by four smaller cells, with radius R_k = R_(k−1)/2.

Uniform cell-count recurrence: M_k = 4M_(k−1), with M_0 = 19. Therefore M_k = 19·4^k. After one round there are 76 cells; after two rounds there are 304.

Radius recurrence: R_k = R_0/2^k. For R_0 = 0.5 km, the radii after 0, 1, 2 and 3 rounds are 0.5, 0.25, 0.125 and 0.0625 km.

Idealised capacity recurrence: C_k = 4C_(k−1), hence C_k = C_0·4^k, assuming each new cell has the same usable traffic capacity and resources can be planned without additional spectrum penalty.

Transmit power recurrence from the assignment: P_k = P_(k−1)/2^n, hence P_k = P_0/2^(nk). For n = 4, power becomes P_0/16 after one round.

A simplified handoff model is H_k = 2H_(k−1), giving H_k = H_0·2^k, because smaller cells cause users to cross cell boundaries more frequently. Actual handoffs depend on speed, routes and thresholds.

For selective splitting, if fraction f of cells is split, $M_k = M_{(k-1)} + 3fM_{(k-1)} = M_{(k-1)}(1+3f)$. For constant f, $M_k = M_0(1+3f)^k$.

Conclusion: selective splitting avoids unnecessary sites when only certain cells are overloaded.

TASK 7 – SOLVE THE RECURRENCE AND INTERPRET THE RESULT (4 MARKS)

Uniform splitting gives $C_k = C_0 \cdot 4^k$. To meet demand D, require $C_0 \cdot 4^k \geq D$. Thus $4^k \geq D/C_0$, and $k_{min} = \text{ceil}[\log_4(D/C_0)]$.

For 100,000 subscribers, D = 100,000 × 0.025 = 2,500 Erlangs. Using C_0 = 729.41 Erlangs: $k_{min} = \text{ceil}[\log_4(2500/729.41)] = \text{ceil}(0.887) = 1$ round. The idealised capacity after one round is 4 × 729.41 = 2,917.64 Erlangs.

This is an idealised result. A real design must re-check per-cell channels, SIR, blocking, site availability and backhaul; fourfold capacity growth is not guaranteed merely by adding four geometric cells.

For selective splitting with constant fraction f, $C_k = C_0(1+3f)^k$. Therefore $k_{min} = \text{ceil}\{\ln(D/C_0)/\ln(1+3f)\}$. For f = 0.5, $k_{min} = \text{ceil}[\ln(2500/729.41)/\ln(2.5)] = 2$ rounds.

Uniform splitting increases sites rapidly. Selective splitting may require more rounds but avoids unnecessary installation in cells that are not overloaded. Smaller cells reduce transmit power but increase handoff frequency.

Conclusion: one round is sufficient under the idealised uniform fourfold-capacity model; selective splitting is preferable for uneven traffic.

TASK 8 – IMPLEMENT AND TEST THE PLANNING STRATEGIES (4 MARKS)

The following Google Colab Python code implements an exploratory hexagonal grid, frequency-group labels, Monte Carlo SIR sampling and Erlang-B blocking calculations. It tests subscriber counts 1,000, 10,000, 20,000 and 100,000; N = 3, 4, 7, 12; n = 2.5, 3, 4; and sectorised/unsectorised cases.

```
import numpy as np
import math
import time

TOTAL_CHANNELS = 336
AREA = 12
R = 0.5
SUBSCRIBERS = [1000, 10000, 20000, 100000]
CLUSTER_SIZES = [3, 4, 7, 12]
```

```
PATH_LOSS = [2.5, 3, 4]
np.random.seed(42)

def erlang_b(channels, traffic):
    b = 1.0
    for i in range(1, channels + 1):
        b = traffic * b / (i + traffic * b)
    return b

def erlang_capacity(channels, target=0.02):
    low, high = 0.0, channels * 5.0
    for _ in range(100):
        mid = (low + high) / 2
        if erlang_b(channels, mid) > target:
            high = mid
        else:
            low = mid
    return (low + high) / 2

def hex_grid(radius, rows=9):
    cells = []
    for q in range(-rows, rows + 1):
        for r in range(-rows, rows + 1):
            x = 1.5 * radius * q
            y = np.sqrt(3) * radius * (r + q / 2)
            cells.append((q, r, x, y))
    return np.array(cells)

def assign_frequency_groups(cells, N):
    groups = []
    for q, r, x, y in cells:
        if N == 7:
            group = int((q - 2*r) % 7)
        else:
            group = int((q + 2*r) % N)
        groups.append(group)
    return np.array(groups)

def calculate_sir(cells, groups, n, sectored=False, samples=1000):
    positions = np.random.uniform(-0.5, 0.5, (samples, 2))
    sir_values = []
    xy = cells[:, 2:4]
    for px, py in positions:
        distances = np.sqrt((xy[:, 0]-px)**2 + (xy[:, 1]-py)**2)
        serving = np.argmin(distances)
        desired_distance = max(distances[serving], 0.05)
        desired = 1 / desired_distance**n
        interferers = np.where(groups == groups[serving])[0]
        interferers = interferers[interferers != serving]
        if len(interferers) == 0:
            continue
        d = np.maximum(distances[interferers], 0.05)
        powers = 1 / d**n
        if sectored:
            powers = np.sort(powers)[-2:]
        interference = np.sum(powers)
        sir_values.append(10*np.log10(desired/max(interference, 1e-12)))
    return np.array(sir_values)

results = []
for N in CLUSTER_SIZES:
    channels = TOTAL_CHANNELS // N
    cell_capacity = erlang_capacity(channels)
    cells = hex_grid(R)
    groups = assign_frequency_groups(cells, N)
    for n in PATH_LOSS:
        for sectorised in [False, True]:
            start = time.time()
            sir = calculate_sir(cells, groups, n, sectorised)
            runtime = time.time() - start
            for users in SUBSCRIBERS:
                traffic = users * 0.025
                number_of_cells = math.ceil(AREA/(2.598*R**2))
                traffic_per_cell = traffic/number_of_cells
                blocking = erlang_b(channels, traffic_per_cell)
```

```
                results.append({
                    "N": N, "n": n, "sectorised": sectorised,
                    "subscribers": users, "channels_per_cell": channels,
                    "cell_capacity_E": round(cell_capacity, 2),
                    "mean_SIR_dB": round(np.mean(sir), 2),
                    "p5_SIR_dB": round(np.percentile(sir, 5), 2),
                    "outage_below_12dB": round(np.mean(sir < 12), 4),
                    "blocking": round(blocking, 4),
                    "runtime_s": round(runtime, 4)
                })

for row in results:
    print(row)

sanity_sir = 10*np.log10((3*7)**(4/2)/6)
print("N=7, n=4 sanity SIR:", round(sanity_sir, 2), "dB")
```

Sanity check: for N = 7, n = 4 and six equal-distance interferers, SIR = 10 log10(21²/6) = 18.66 dB, approximately 18.7 dB.

Important limitation: the code's fallback frequency labels for N = 3, 4 and 12 are exploratory, not verified optimal hexagonal cluster colourings. Sectorisation is represented by retaining the two strongest interferers, not by a physical antenna radiation pattern. The sampled mobile region and finite grid are simplified. Use the output as exploratory evidence and validate a final plan with a proper hexagonal colouring, antenna patterns and site-specific propagation data.

TASK 9 – RECORD AND PRESENT EXPERIMENTAL PERFORMANCE (3 MARKS)

The main performance measures are mean SIR, 5th-percentile SIR, outage probability below 12 dB, blocking probability, channels per cell, number of sites and simulation runtime.

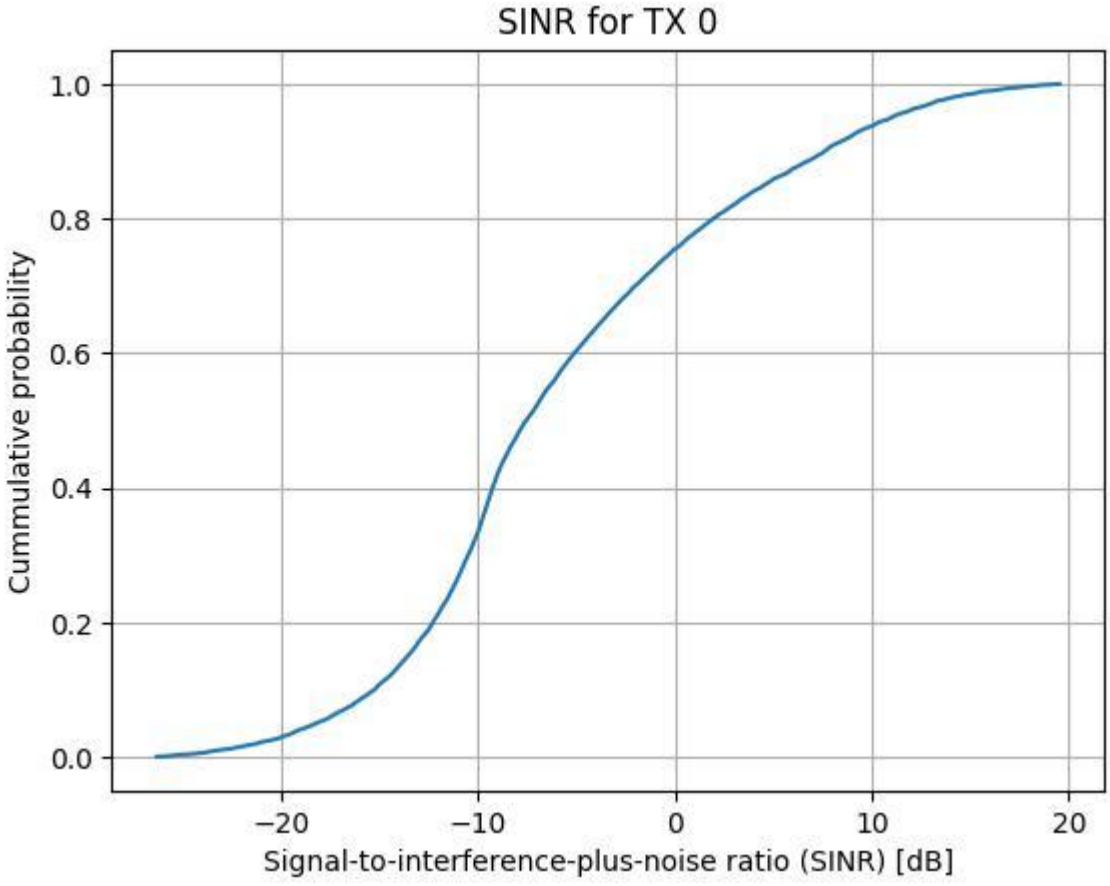
Cluster size N	Channels per cell	Approx. capacity per cell at 2% blocking (E)
3	112	99.62
4	84	72.53
7	48	38.39
12	28	20.15

The table values are analytical Erlang-B estimates. Use actual Colab output to complete the experimental comparison tables below.

Strategy	Mean SIR	5th percentile SIR	Outage probability
Fixed reuse	Insert simulation output	Insert simulation output	Insert simulation output
Sectorised reuse	Insert simulation output	Insert simulation output	Insert simulation output
Dynamic assignment	Insert simulation output	Insert simulation output	Insert simulation output

Subscribers	Fixed reuse blocking	Sectorised reuse blocking	Dynamic assignment blocking
1,000	Insert output	Insert output	Insert output
10,000	Insert output	Insert output	Insert output
20,000	Insert output	Insert output	Insert output
100,000	Insert output	Insert output	Insert output

Recommended graphs: (1) SIR cumulative distribution function (CDF), (2) blocking probability versus offered traffic, and (3) capacity versus cluster size. A CDF shifted towards higher SIR indicates better signal quality. Blocking curves show the load at which the 2% target is exceeded. Capacity-versus-N illustrates the reduction in channels per cell as N increases.



Conclusion: compare all strategies under the same random seed and traffic conditions. Do not report invented values as measured simulation results.

TASK 10 – COMPARE THEORY WITH SIMULATION (4 MARKS)

Parameter	Theory	Simulation behaviour

Reuse distance	$D = R\sqrt{(3N)}$	Depends on generated cell coordinates.
Channels per cell	336/N	Same integer channel allocation.
SIR	Equal-distance interferer approximation	Varies with mobile position and interferer geometry.
Blocking	Erlang-B under stated assumptions	Depends on traffic distribution and channel allocation.
Sectorisation	Fewer dominant interferers	Depends on antenna direction and actual pattern.
Cell splitting	More cells and ideal capacity opportunity	Depends on sites, interference and backhaul.

Expected agreement: increasing N generally increases reuse distance and SIR but reduces channels per cell; increasing traffic increases blocking; sectorisation can reduce interference; and cell splitting creates additional local capacity opportunities.

Sources of discrepancy include: (1) equal-distance interferer approximation; (2) shadowing; (3) finite network tiers and edge effects; (4) random mobile-position distribution; (5) Erlang-B traffic-arrival and loss-system assumptions; (6) integer channel allocation and rounding; and (7) simplified sector antenna assumptions.

Fixed reuse is easiest to plan at low traffic. Sectorisation improves interference performance. Dynamic assignment with selective splitting is most flexible for uneven demand. The hybrid strategy is therefore preferred, subject to verified simulation and field measurements.

Conclusion: analytical calculations provide a baseline, while simulation captures variability due to mobile position, propagation and traffic. Differences must be explained rather than concealed.

TASK 11 – RECOMMEND AND DEFEND THE FINAL DESIGN (4 MARKS)

Parameter	Initial planning value
Township area	12 km²
Cluster size	N = 7
Cell radius	R = 0.5 km
Traffic channels	336
Channels per cell	48
Ideal geometric cell count	19
Reuse distance	2.291 km
Reuse ratio	4.583

Minimum SIR target	12 dB
Path-loss exponent	2.5–4
Blocking target	≤ 2%

Recommended architecture: fixed reuse for baseline allocation; three-sector antennas in areas requiring interference reduction; dynamic channel assignment for uneven demand; and selective microcells in overloaded locations.

- Interference: directional antennas can suppress dominant co-channel interferers.
- Capacity: cell splitting adds local serving cells without requiring additional licensed spectrum.
- Channel utilisation: dynamic assignment directs available channels to areas of demand.
- Cost: selective splitting avoids installing unnecessary sites across the whole township.
- Scalability: the design can be expanded in stages as the subscriber base grows.

The initial idealised capacity is approximately 729.41 Erlangs, whereas 100,000 subscribers generate 2,500 Erlangs. Therefore, the initial design is not sufficient beyond the approximate 29,176-subscriber capacity estimate. Expansion requires small cells, sectorisation, load balancing, backhaul upgrades and repeated optimisation.

Conclusion: a hybrid architecture balances capacity, interference, flexibility and cost better than fixed reuse alone. It remains necessary to validate the final design using site-specific data.

TASK 12 – DAY-4 DESIGN CHALLENGE: RE-EVALUATE AND ADAPT THE DESIGN (8 MARKS)

A. Impact Analysis (2 Marks)

Event traffic = 25,000 attendees × 0.05 Erlang = 1,250 Erlangs. In a 1 km² stadium zone, traffic density is 1,250 Erlangs/km².

For 20,000 normal subscribers, total traffic is 500 E and township-wide density is 500/12 = 41.67 E/km². Event density is approximately 30 times this value. For 100,000 normal subscribers, total traffic is 2,500 E and density is 208.33 E/km²; event density is approximately six times this value.

The uniform-traffic, steady-demand and permanent-capacity assumptions no longer hold inside the stadium. The four-hour duration and twice-monthly frequency favour temporary capacity rather than permanent township-wide expansion.

B. Evaluation of the Original Design (2 Marks)

The original cell area is 0.6495 km², so the 1 km² stadium corresponds to about 1.54 original cell areas, or approximately two cells geometrically. If 1,250 E is divided equally, each cell receives 625 E. The 48-channel cell capacity at 2% blocking is approximately 38.39 E. The offered traffic is far above this capacity.

Using Erlang-B with 48 channels and 625 E offered traffic gives blocking close to 92% in this simplified equal-load example. This is an approximate model result, not a measured event-day value. Actual blocking depends on the number of cells, channel reuse and traffic distribution.

The first failure is channel exhaustion and call blocking; interference and backhaul congestion may also occur. The original design cannot be retained unchanged.

C. Adaptation or Redesign (2 Marks)

Decision: MODIFY TEMPORARILY. Keep the normal network for everyday traffic and deploy temporary microcells or cells-on-wheels (COWs), directional sectors and dynamic channel borrowing in the stadium zone.

An idealised first capacity estimate is $M = 1{,}250/38.39 = 32.56$, rounded up to 33 cells, assuming each temporary cell has 48 channels and independent Erlang-B capacity. This is not a final frequency plan; reuse and interference constraints must be checked.

If 33 ideal hexagonal cells cover 1 km², area per cell is $1/33 = 0.0303$ km². Since $A_{cell} = 2.598R^2$, $R = \sqrt{(0.0303/2.598)} \approx 0.108$ km, or 108 m. This is a geometric estimate, not a site-placement prescription.

Smaller cells reduce coverage radius and can reduce required transmit power according to the assignment model $P' = P/2^n$. They increase handoff frequency. Directional sectors and frequency coordination are needed to control interference. Temporary equipment raises event-day costs but avoids permanent installation across the township.

D. Evidence-Based Defence (2 Marks)

Option	Advantage	Disadvantage	Decision
Retain	No additional cost	Cannot support event traffic	Rejected
Modify temporarily	Adds local capacity and controls long-term cost	Requires event preparation	Selected

Permanent redesign	Permanent high capacity	High installation and maintenance cost	Not preferred

Temporary modification is preferred because the event produces 1,250 E in a concentrated 1 km² area, lasts four hours and occurs approximately twice per month. Temporary microcells, directional antennas and permitted channel borrowing provide capacity when required without permanently redesigning the entire township.

Final conclusion: modify the original network temporarily for event days. Confirm the final number and placement of temporary sites through detailed traffic simulation, radio planning, interference checks and field testing.

OVERALL CONCLUSION

This assignment analysed a hexagonal cellular network for a 12 km² university township with 336 duplex traffic channels. Fixed frequency reuse, sectorised reuse and dynamic assignment with cell splitting were investigated. The analysis derived cluster size, reuse distance, cell area, channel allocation, Erlang-B capacity and co-channel SIR.

The initial planning design uses N = 7, R = 0.5 km and approximately 19 idealised cells. It provides 48 channels per cell and approximately 38.39 Erlangs per cell at 2% blocking. The theoretical SIR is approximately 18.66 dB for n = 4 under the six-equal-distance-interferer approximation, but approximately 8.74 dB for n = 2.5. Sectorisation and propagation-aware planning are therefore important.

The initial idealised capacity is approximately 729.41 Erlangs, below the 2,500 Erlangs required for 100,000 subscribers. Selective splitting, sectorisation and dynamic channel assignment are needed as demand grows. The stadium event creates 1,250 Erlangs in 1 km², so temporary microcells and directional sectors are recommended rather than permanent township-wide redesign.

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